Joint fit of common and decorrelating directions

The plots show expected constraints on $C_{Ha}^{(3)}$ and n_7 from normalized longitudinal WZ and Wh rates. The observables $S=(\mu_Z+\mu_h)/2-1$ and $D=(\mu_Z-\mu_h)/2$ measure their average shift and difference, respectively; μ denotes the rate divided by its SM prediction.

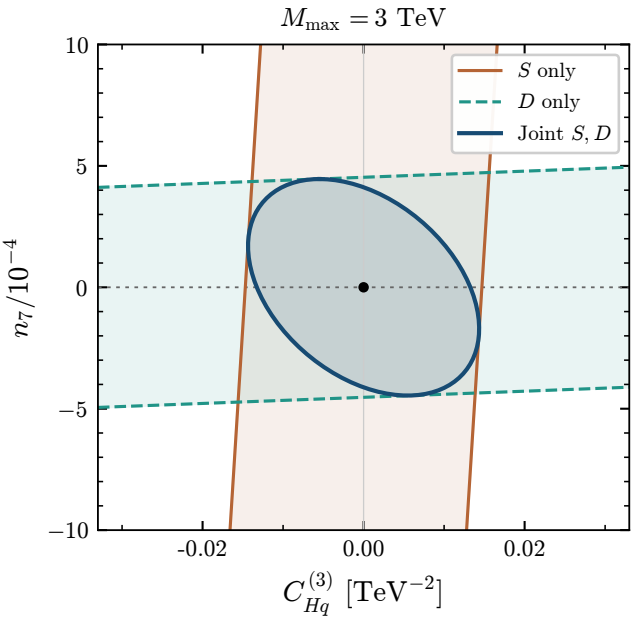


FIG. 1. S-only (orange), D-only (green dashed), and joint S, D (blue) constraints for $M < M_{\max} = 3$ TeV. Both coefficients float. The single-observable bands extend beyond the frame.

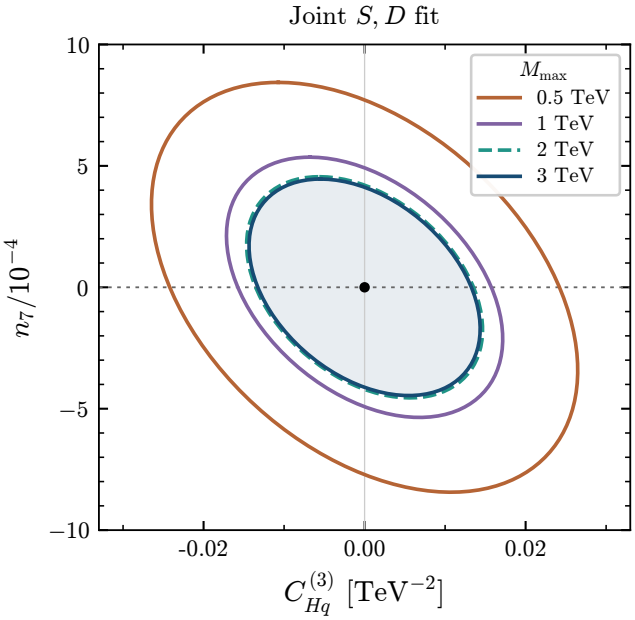


FIG. 2. Joint S, D constraints for $M_{\max} = 0.5, 1, 2, 3$ TeV. Each contour uses a separate cumulative mass-selected sample, with fixed transverse-momentum bins and fractional errors.

In Fig. 1, the nearly vertical S-only band mainly constrains $C_{Hq}^{(3)}$, while the nearly horizontal D-only band mainly constrains n_7. Combining them, including their covariance, produces a compact, tilted ellipse: the two observables constrain complementary directions.

In Fig. 2, the contours shrink as the upper mass cut increases. The largest improvement occurs between 0.5 and 2 TeV; the 2 and 3 TeV contours nearly overlap. From 0.5 to 3 TeV, the profiled uncertainties improve by factors of about 1.9 for n_7 and 1.8 for $C_{Hq}^{(3)}$, with only a further 2% improvement from 2 to 3 TeV.

All boundaries correspond to $\Delta\chi^2 = 2.30$ (68.3% joint confidence for two parameters). The black point marks the SM; the horizontal dotted line, $n_7 = 0$, is the selected SMEFT direction. Every contour is centered on the SM, so these are expected sensitivities rather than evidence for a nonzero deformation.

Benchmark: $\sqrt{s} = 14$ TeV, nCTEQ15, interference-order templates, and p_T bins of 100-200 and 200-1500 GeV. Fractional channel errors are $\epsilon_Z = (0.088, 0.091)$ and $\epsilon_h = (0.15, 0.15)$, with uncorrelated channel and bin measurements. The mass scan holds these errors fixed.